

Комп практикум 5 лаба

Отчет по лабораторной работе

Веб-приложение на Flask (GET и POST запросы)

1. Инструментарий

- Python 3.14.3
- Flask 3.1.3
- curl для тестирования запросов
- Браузер для отображения формы

2. Установка Flask

```
# Снятие изоляции и установка
pip3 install flask --break-system-packages
# Настройка pip.conf для постоянной работы
mkdir -p ~/.config/pip
nano ~/.config/pip/pip.conf # добавлено break-system-packages = true
```

```
To install flask, run:
  brew install flask
nerdysnake6@MacBook-Air-Kalinin ~ % flask --version

zsh: command not found: flask
nerdysnake6@MacBook-Air-Kalinin ~ % pip3 install flask --break-system-packages
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
```

3. Код приложения (app.py)

```
from flask import Flask, request, render_template_string, jsonify

app = Flask(__name__)

HTML_FORM = """
<!DOCTYPE html>
<html>
<head>
  <title>Flask App</title>
  <style>
    body { font-family: Arial; padding: 20px; background: #f0f0f0; }
    .container { max-width: 600px; margin: 0 auto; background: white;
padding: 20px; border-radius: 10px; }
```

```

        input, textarea { width: 100%; padding: 8px; margin: 5px 0; }
        button { background: #007bff; color: white; border: 0; padding:
10px; cursor: pointer; }
        pre { background: #eee; padding: 10px; }
    </style>
</head>
<body>
    <div class="container">
        <h1>Flask App</h1>
        <p><b>GET /</b> – эта форма</p>
        <p><b>GET /api</b> – JSON ответ</p>
        <p><b>POST /api</b> – принять данные</p>

        <h3>Отправить POST запрос</h3>
        <form method="post" action="/api">
            <input type="text" name="name" placeholder="Имя" required>
            <input type="email" name="email" placeholder="Email" required>
            <textarea name="message" placeholder="Сообщение"></textarea>
            <button type="submit">Отправить</button>
        </form>

        <h3>Последний ответ:</h3>
        <pre>{{ response }}</pre>
    </div>
</body>
</html>
"""

```

```

@app.route('/', methods=['GET'])
def home():
    return render_template_string(HTML_FORM, response="Пока ничего нет")

```

```

@app.route('/api', methods=['GET'])
def api_get():
    return jsonify({
        "method": "GET",
        "message": "Это GET запрос",
        "args": dict(request.args)
    })

```

```

@app.route('/api', methods=['POST'])
def api_post():
    data = {
        "method": "POST",
        "form": dict(request.form),
        "message": "Данные получены"
    }

```

```

}

html_response = f"""
Метод: POST<br>
Имя: {request.form.get('name', '')}<br>
Email: {request.form.get('email', '')}<br>
Сообщение: {request.form.get('message', '')}
"""

if request.headers.get('Accept') == 'application/json':
    return jsonify(data)
else:
    return render_template_string(HTML_FORM, response=html_response)

if __name__ == '__main__':
    app.run(debug=True, port=5000)

```

4. Запуск приложения

```
flask-app — Python · Python app.py — 101x44
Downloading werkzeug-3.1.6-py3-none-any.whl (225 kB)
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 click-8.3.1 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsa
fe-3.0.3 werkzeug-3.1.6
nerdysnake6@MacBook-Air-Kalinin ~ % mkdir -p ~/.config/pip

nerdysnake6@MacBook-Air-Kalinin ~ % nano ~/.config/pip/pip.conf

nerdysnake6@MacBook-Air-Kalinin ~ % pip3 install flask

Requirement already satisfied: flask in /opt/homebrew/lib/python3.14/site-packages (3.1.3)
Requirement already satisfied: blinker>=1.9.0 in /opt/homebrew/lib/python3.14/site-packages (from fla
sk) (1.9.0)
Requirement already satisfied: click>=8.1.3 in /opt/homebrew/lib/python3.14/site-packages (from flask
) (8.3.1)
Requirement already satisfied: itsdangerous>=2.2.0 in /opt/homebrew/lib/python3.14/site-packages (fro
m flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in /opt/homebrew/lib/python3.14/site-packages (from flas
k) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in /opt/homebrew/lib/python3.14/site-packages (from
flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in /opt/homebrew/lib/python3.14/site-packages (from fl
ask) (3.1.6)
nerdysnake6@MacBook-Air-Kalinin ~ % flask --version

Python 3.14.3
Flask 3.1.3
Werkzeug 3.1.6
nerdysnake6@MacBook-Air-Kalinin ~ % mkdir ~/flask-app && cd ~/flask-app
nerdysnake6@MacBook-Air-Kalinin flask-app % nano app.py
nerdysnake6@MacBook-Air-Kalinin flask-app % python3 app.py
* Serving Flask app 'app'
[ * Debug mode: on
[WARNING: This is a development server. Do not use it in a production deployment. Use a production WSG
I server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 114-834-344
127.0.0.1 - - [12/Mar/2026 15:46:20] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [12/Mar/2026 15:46:20] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [12/Mar/2026 15:47:39] "POST /api HTTP/1.1" 200 -
```

5. Демонстрация работы

Flask App

GET / - эта форма

GET /api - JSON ответ

POST /api - принять данные

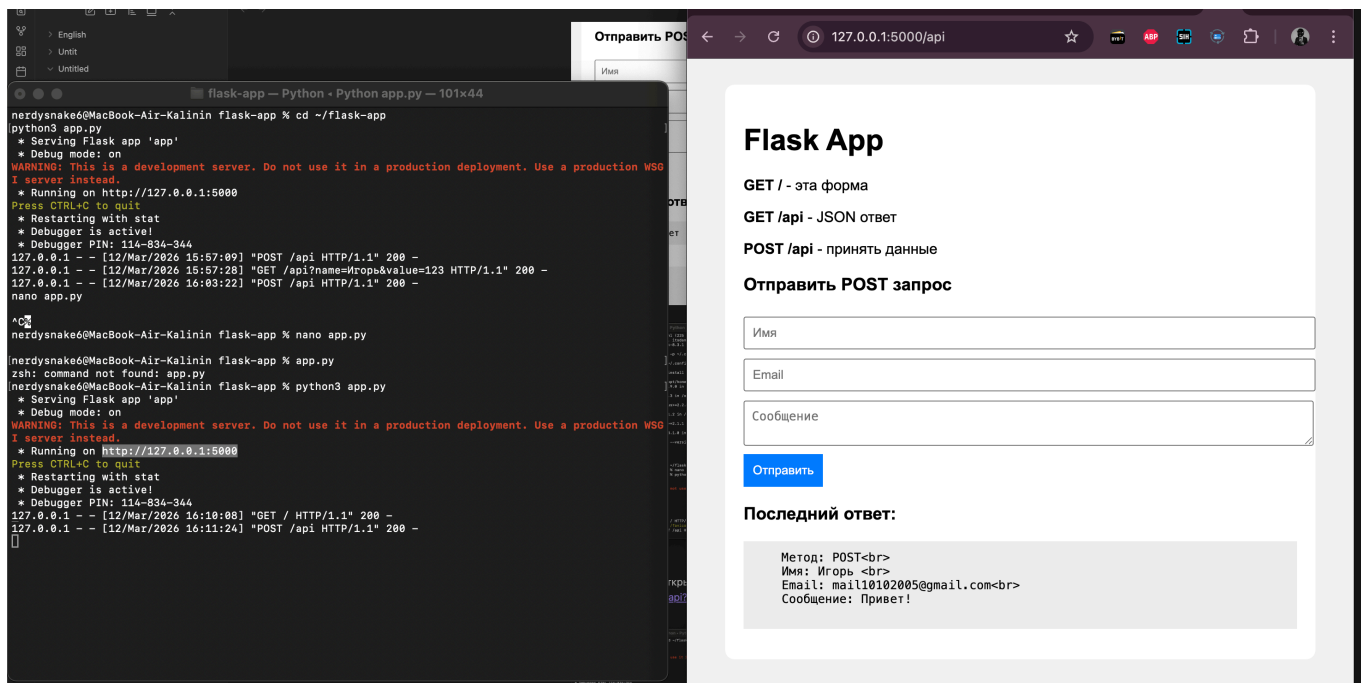
Отправить POST запрос

Отправить

Последний ответ:

Пока ничего нет

Post запрос



Get запрос

Для get запроса я открывал ссылку

<http://127.0.0.1:5000/api?name=Игорь&value=123>

